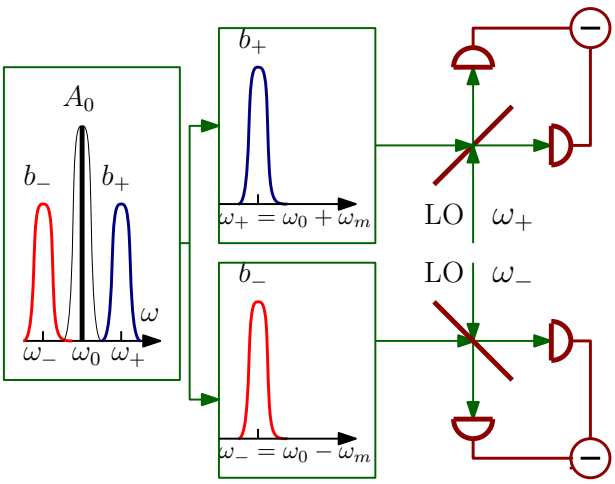

Figure 3. A scheme of the measurement. Quadrature components of the output modes $\omega_\pm$ are measured separately by balanced homodyne detectors with corresponding optimal local oscillators having frequencies $\omega_\pm$. The signal is inferred by processing of the linear combination of the measured results. Essentially, the linear combination in frequency domain should have complex frequency dependent coefficients.

The Fourier transform of operators, for example, $\hat{a}_\pm$ is defined as follows

$$\hat{a}_\pm(t) = \int_{-\infty}^{\infty} a_\pm(\Omega)\, e^{-i\Omega t}\, \frac{d\Omega}{2\pi}. \tag{3.7}$$

For operators $a_\pm(\Omega)$ the following commutators and correlators are valid:

$$\left[a_\pm(\Omega), a_\pm^\dagger(\Omega')\right] = 2\pi\, \delta(\Omega - \Omega'), \tag{3.8}$$

$$\left\langle a_\pm(\Omega) a_\pm^\dagger(\Omega') \right\rangle = 2\pi\, \delta(\Omega - \Omega') \tag{3.9}$$

Similar expressions can be written for the other noise operators ($\hat{e}_\pm$, $\hat{q}$).

We assume that the signal force is a resonant square pulse acting during time interval τ ($\omega_m \tau \gg 1$):

$$F_S(t) = F_{s0} \sin(\omega_m t + \psi_f) = \tag{3.10}$$
$$= i\left(F_s(t) e^{-i\omega_m t} - F_s^*(t) e^{i\omega_m t}\right), \quad -\frac{\tau}{2} < t < \frac{\tau}{2},$$

$$f_s(\Omega) = \frac{F_s(\Omega)}{\sqrt{2\hbar\omega_m m}}, \tag{3.11}$$

$$f_{s0}(\Omega) = \frac{F_{s0}(\Omega)}{\sqrt{2\hbar\omega_m m}} = 2 f_s(\Omega). \tag{3.12}$$

where $F_s(\Omega) \neq F_s^*(-\Omega)$ is the Fourier amplitude of $F_s(t)$.

Let introduce quadrature amplitudes of amplitude and phase

$$a_{\pm a} = \frac{a_\pm(\Omega) + a_\pm^\dagger(-\Omega)}{\sqrt{2}}, \tag{3.13a}$$

$$a_{\pm\phi} = \frac{a_\pm(\Omega) - a_\pm^\dagger(-\Omega)}{i\sqrt{2}}. \tag{3.13b}$$

Analysis shows that sum $(c_{+a} + c_{-a})$ does not contain information on the mechanical motion but contains the back action term. At the same time difference $(c_{+\phi} - c_{-\phi})$ does not contain information on mechanical motion (term $\sim d_\phi$), but is responsible on back action (see detailes in Appendix B). So it should be useful to introduce sum and difference of the quadratures

$$g_{a\pm} = \frac{c_{+a} \pm c_{-a}}{\sqrt{2}}, \quad g_{\phi\pm} = \frac{c_{+\phi} \pm c_{-\phi}}{\sqrt{2}}, \tag{3.14a}$$

$$\alpha_{a\pm} = \frac{a_{+a} \pm a_{-a}}{\sqrt{2}}, \quad \alpha_{\phi\pm} = \frac{a_{+\phi} \pm a_{-\phi}}{\sqrt{2}}, \tag{3.14b}$$

$$\beta_{a\pm} = \frac{b_{+a} \pm b_{-a}}{\sqrt{2}}, \quad \beta_{\phi\pm} = \frac{b_{+\phi} \pm b_{-\phi}}{\sqrt{2}}, \tag{3.14c}$$

$$\epsilon_{a\pm} = \frac{e_{+a} \pm e_{-a}}{\sqrt{2}}, \quad \epsilon_{\phi\pm} = \frac{e_{+\phi} \pm e_{-\phi}}{\sqrt{2}} \tag{3.14d}$$

Recall we can measure output quadratures from modes $\omega_\pm$ separately, so it is naturally to take sums and differences of quadratures.

A. Measurement of amplitude quadratures

Recall we measure output quadratures of modes $\omega_\pm$ *separately* using balanced homodyne detectors with local oscillators frequencies $\omega_\pm$. Let we find output amplitude quadratures. It is convenient to present the solution of set (B7a, B7b, B7c) in Appendix B for sum and difference $\beta_{a\pm}$ of the output amplitude quadratures:

$$\beta_{a+} = \xi_+ \left\{\alpha_{a+} + \frac{\mu_+}{\xi_+}\epsilon_{a+}\right\}, \tag{3.15a}$$

$$\xi_+ = \frac{\gamma_0 - \gamma_e + \kappa + i\Omega}{\gamma_0 + \gamma_e - \kappa - i\Omega}, \quad \xi_- = \frac{\gamma_0 - \gamma_e - \kappa + i\Omega}{\gamma_0 + \gamma_e + \kappa - i\Omega} \tag{3.15b}$$

$$\mu_\pm = \frac{2\sqrt{\gamma_0\gamma_e}}{\gamma_0 + \gamma_e \mp \kappa - i\Omega}, \tag{3.15c}$$

$$\beta_{a-} = \xi_- \left[\alpha_{a-} + \frac{\mu_-}{\xi_-}\epsilon_{a-} - \right. \tag{3.15d}$$

$$\left. - \frac{\mathcal{K}}{\gamma_m - i\Omega}\left\{\alpha_{a+} + \sqrt{\frac{\gamma_e}{\gamma_0}}\,\epsilon_{a+}\right\}\right] - \tag{3.15e}$$

$$- \frac{\sqrt{\xi_-\mathcal{K}}}{\gamma_m - i\Omega}\left(\sqrt{2\gamma_m} q_a + f_{s\,a}\right), \tag{3.15f}$$

$$\mathcal{K} \equiv \frac{4\gamma_0\, \eta^2 C_0^2}{\gamma_0^2 - (\kappa + \gamma_e - i\Omega)^2} \tag{3.15g}$$

It is useful to write dimensionless power $\mathcal{K}$ in form

$$\mathcal{K} = \mathcal{K}_0\, \frac{\gamma(\gamma_0 - \gamma_e)}{\gamma_0^2 - (\kappa + \gamma_e - i\Omega)^2}, \tag{3.16}$$

$$\mathcal{K}_0 = \frac{4\gamma_0\omega_0 P_{in}}{m\omega_m L^2\gamma^2(\gamma_0 - \gamma_e)}, \tag{3.17}$$

where L is cavity length, P_{in} — input light power.

The result of parametric (internal) gain κ is two-photon (nondegenerate) unsqueezing of sum quadrature

β_{a+} (3.15a), whereas part (3.15d) of β_{a-} is squeezed. It is obvious if put $\Omega = 0$ and account (2.1). The squeezing increases as κ grows, but parametric gain is restricted by condition of stability.

$$\kappa \leq \gamma_0 + \gamma_e \tag{3.18}$$

The two photon (nondegenerate) *external* squeezing is described by squeezing (or unsqueezing) of quadratures $\alpha_\pm$ of input field.

As expected, in Eq. (3.15e) the back action term is proportional to the normalized mean energy $\mathcal{K}$, stored in cavity. However, this term can be excluded by the post processing. One can measure *both* β_{a+} and β_{a-} simultaneously and (after proper filtration) subtract term in figure brackets in β_{a+} (3.15a) from term in figure brackets in β_{a-} (3.15e) to remove the back action. However we can not subtract back action completely, we can subtract term $\sim \alpha_{a+}$ completely but term $\sim \epsilon_{a+}$ survives. It means that we can measure combination

$$\tilde{\beta}_{a-} = \beta_{a-} + \frac{\xi_- \mathcal{K}}{\gamma_m - i\Omega}\frac{\beta_{a+}}{\xi_+} = \tag{3.19a}$$

$$= \xi_- \left[\alpha_{a-} + \frac{\mu_-}{\xi_-}\epsilon_{a-} + \right. \tag{3.19b}$$

$$\left. + \frac{\mathcal{K}}{\gamma_m - i\Omega}\left\{\sqrt{\frac{\gamma_e}{\gamma_0}}\frac{\epsilon_{a+}}{\xi_+^*}\right\}\right] - \tag{3.19c}$$

$$- \frac{\sqrt{\xi_- \mathcal{K}}}{\gamma_m - i\Omega}\left(\sqrt{2\gamma_m} q_a + f_{s\,a}\right) \tag{3.19d}$$

Essentially, the coefficient needed for suppression of the back action is complex, it depends on the spectral frequency Ω. It means post processing filtration.

The term (3.19c) describes residual back action and it is loss fluctuations ϵ_{a+} are responsible for it. Obviously that residual back action restricts sensitivity but due to smallness of optical loss ($\gamma_e \ll \gamma_0$) SQL can be surpassed.

We find the force detection condition using single-sided power spectral density $S_f(\Omega)$ for signal force (3.10). Assuming that the detection limit corresponds to the signal-to-noise ratio exceeding unity we obtain recalculating (3.15d) to signal force:

$$\tilde{\beta}_{a-}\frac{(\gamma_m - i\Omega)}{\sqrt{\xi_- \mathcal{K}}} = -f_{s0} - \sqrt{2\gamma_m}\, q_a + \tag{3.20a}$$

$$+ \frac{\sqrt{\xi_-}\,(\gamma_m - i\Omega)}{\sqrt{\mathcal{K}}}\left(\alpha_{a-} + \frac{\mu_-}{\xi_-}\epsilon_{a-}\right) - \tag{3.20b}$$

$$- \sqrt{\xi_- \mathcal{K}}\left\{\sqrt{\frac{\gamma_e}{\gamma_0}}\frac{\epsilon_{a+}}{\xi_+^*}\right\} \tag{3.20c}$$

$$f_{s0} \geq \sqrt{S_f(\Omega)\cdot\frac{\Delta\Omega}{2\pi}}, \tag{3.20d}$$

where $\Delta\Omega \simeq 2\pi/\tau$ is defined by time τ of signal force action.

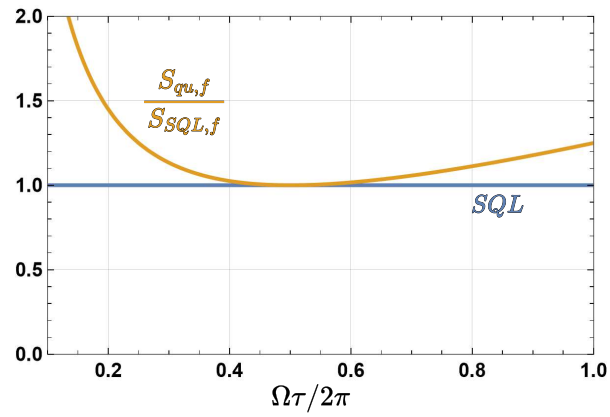

Figure 4. Plot of spectral density $S_{qu,f}$ (3.24) and $S_{SQL,f}$ (3.25) as function of spectral frequency Ω, dimensionless power $\mathcal{K}_0 = \pi/\tau$ (3.17), τ is time of signal force action which is chosen to be $\tau = 0.28$ msec.

1. *No squeezing, no optical losses, no subtraction of back action*

Let consider simplest case without squeezing ($\kappa = 0$) and losses ($\gamma_e = 0$), measurement of β_{a-} (3.15d – 3.15f) *only* (no subtraction of back action). Using (B2, B4) we derive

$$\gamma_e = 0, \quad \kappa = 0, \quad \Rightarrow \quad |\xi_\pm| = 1,\ |\mu_\pm| = 0 \tag{3.21}$$

$$S_f^{\beta_{a-}}(\Omega) = 2\gamma_m\big(2n_T + 1\big) + S_{qu,f}, \tag{3.22}$$

$$S_{qu,f} \geq S_{SQL,f}, \tag{3.23}$$

$$S_{qu,f} = \frac{\gamma_m^2 + \Omega^2}{\mathcal{K}} + \mathcal{K}, \tag{3.24}$$

$$S_{SQL,f} = 2\sqrt{\gamma_m^2 + \Omega^2} \tag{3.25}$$

The sensitivity is restricted by SQL. Here we assume that normalized pump $\mathcal{K}$ (3.16) practically does not depend on spectral frequency Ω due to condition (2.1) and $\Omega \ll \gamma_0$. Strictly speaking, formula (3.25) is a kind of mind game — minimization is taken at optimal $\mathcal{K}$, depending of spectral frequency Ω, it can not be realized in experiment. However, more accurate consideration gives result only slightly differing from result with usage (3.25) (see details in Appendix C). So below we use (3.25) for SQL characterization. The examples of plots of $S_{qu,f}$ and $S_{SQL,f}$ are presented on Fig 4, where we put $\mathcal{K}_0 = \pi/\tau$ (3.17), τ is time of signal force action. For parameters listed in Table I it corresponds to $P_{in} \simeq 10^{-2}$ W.

The thermal noise (term $\sim \gamma_m$ in (3.22) prevents detection of signal force in any opto-mechanical device. Below we assume that Braginsky condition [13, 14] of smallness of thermal noise as compared with SQL

$$B = \frac{n_T \omega_m \tau}{Q} \ll 1 \tag{3.26}$$

is fulfilled (Q is a mechanical quality factor). The main requirement for this is large ring down time γ_m^{-1} and fast

Table I. Parameters of a mechanical oscillator (SiN membrane) and optical cavity, used for estimates.

Membrane		
Mass, m	50	10^{-9} g
Frequency, $\omega_m/2\pi$	350	10^3 Hz
Quality factor $Q = \frac{\omega_m}{2\gamma_m}$	10^8	
Temperature, T	20	K°
Thermal phonons number, n_T	$1.2 \cdot 10^6$	
Time of signal force $\tau = 10 \cdot \frac{2\pi}{\omega_m}$	$28 \cdot 10^{-6}$	sec
Cavity		
Length of cavity	10	cm
Power transmittance of input mirror, T^2	$3 \cdot 10^{-4}$	
Power losses (in cavity), ϵ^2	10^{-6}	
Bandwidth , $(\gamma_0 + \gamma_e)/2\pi$	36	10^3 Hz
Wave length, $\lambda = 2\pi c/\omega_0$	1.55	10^{-6} m
Input power P_{in}	10	10^{-3} W

interrogation time τ, i.e. $\gamma_m \tau \ll 1$. Note, for parameters listed in Table I factor $B \simeq 0.75$ and only for $Q \simeq 10^9$ condition (3.26) is realized.

2. *No squeezing, no optical losses, but subtraction of back action*

We again consider simplest case without squeezing and losses ($\kappa = \gamma_e = 0$), but with subtraction of back action via post processing. It means the measurement of combination $\tilde{\beta}_{a-}$ (3.19). Then the spectral density is *not* limited by SQL:

$$\gamma_e = 0, \quad \kappa = 0, \quad \Rightarrow \quad |\xi_\pm| = 1, \ |\mu_\pm| = 0 \tag{3.27}$$

$$S_f^{\tilde{\beta}_{a-}}(\Omega) = 2\gamma_m\big(2n_T + 1\big) + \frac{\gamma_m^2 + \Omega^2}{\mathcal{K}} \tag{3.28}$$

Here the first term describes thermal noise and the second one stands for the quantum measurement noise (shot noise) decreasing with the power increase. The back action term is excluded *completely* [43].

3. *No external squeezing, nondegenerate squeezing inside cavity, account of losses, no back action subtraction*

Here we assume that there are two-photon (nondegenerate) internal squeezing and optical losses ($\kappa \neq 0, \gamma_e \neq 0$), input fluctuations fields ($\alpha_{a\pm}$, $\epsilon_\pm$) are in vacuum state (no external squeezing). But back action subtraction is not applied. Using (3.15d, 3.15e, 3.15f) we derive power spectral density for this case

$$\gamma_e \neq 0, \quad \kappa \neq 0, \quad \text{vacuum input} \ \Rightarrow \tag{3.29a}$$

$$S_f^{\beta_{a-}}(\Omega) = 2\gamma_m\big(2n_T + 1\big) + \tag{3.29b}$$

$$+ \frac{\gamma_m^2 + \Omega^2}{|\mathcal{K}|}\left(|\xi_-| + \frac{|\mu_-|^2}{|\xi_-|}\right) + \tag{3.29c}$$

$$+ |\xi_- \mathcal{K}|\left\{1 + \frac{\gamma_e}{\gamma_0}\right\} \tag{3.29d}$$

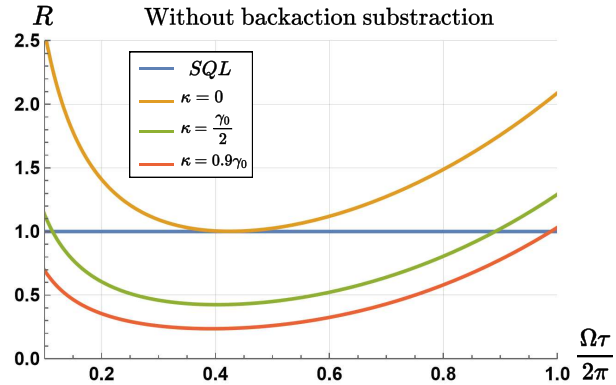

Figure 5. Plots of ratio $R = S_f^{\beta_{a-}}/S_{SQL,f}$ of spectral densities (3.29, 3.25) as function of spectral frequency Ω at condition (3.29a) and $\gamma_m = 0$. Dimensionless power $\mathcal{K}_0 = \pi/\tau$ (3.17), τ is time of signal force action. Other parameters are taken from Table I.

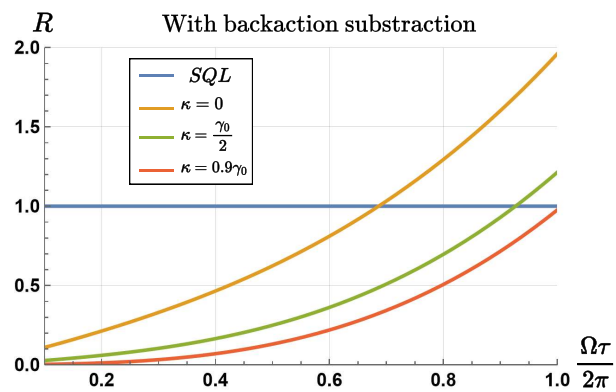

Figure 6. Plots of ratio $R = S_f^{\tilde{\beta}_{a-}}/S_{SQL,f}$ of spectral densities (3.30, 3.25) as function of spectral frequency Ω at condition (3.30a) and $\gamma_m = 0$. Dimensionless power $\mathcal{K}_0 = \pi/\tau$ (3.17), τ is time of signal force action. Other parameters are taken from Table I.

Here first term (3.29b) describes thermal noise, second one (3.29c) — measurement error and last term (3.29d) — back action.

We present on Fig. 5 the plots of spectral density (3.29), normalized to SQL (i.e. $R = S_f^{\beta_{a-}}/S_{SQL,f}$) for different parameters of squeezing. For case without squeezing $\kappa = 0$ we have usual SQL, but for squeezing $\kappa = \gamma_0/2$, $0.9\,\gamma_0$ we have considerable gain of sensitivity.

Recall, system is stable at condition (3.18).